

Problem 1.

a) Express the following complex number in the form of $x + yi$ for $x, y \in \mathbf{R}$.

$$\frac{1 + 2i}{5 + i}.$$

b) Express the following complex number in the form of $re^{i\theta}$ for $r > 0, \theta \in [0, 2\pi)$.

$$\frac{2 + i}{\sqrt{10}}.$$

c) What complex numbers satisfy the following equation?

$$z^n = 1$$

Here, n is a positive number.

d) Suppose we have a polynomial

$$f(z) = \sum_{i=0}^n a_i z^i$$

where $a_i \in \mathbf{C}$. Then, show that if $f(w) = 0$ for some complex number w then

$$f(z) = (z - w)g(z) \quad \text{for all } z \in \mathbf{C}.$$

Here, $g(z)$ is some polynomial of degree $n - 1$.

Problem 2. Let V and W be vector spaces over $\mathbf{F}$ and $T : V \rightarrow W$.

a) Write the definition that T is linear.

b) Let $V, W = R[x]$ (polynomials with real coefficients) and $\mathbf{F} = \mathbf{R}$. Show that the differentiation

$$D : p(x) \mapsto p'(x)$$

is linear.

c) Show that T is injective if and only if $\text{null}(T) = \{0\}$.

d) Suppose $\dim V$ is finite and $\text{null}(T) = \{0\}$. Then, prove that $\dim(V) = \dim(\text{range}(T))$.
Hint: suppose $\{v_1, \dots, v_n\}$ is a basis for V .

Problem 3. Let V be the n dimensional vector space over $\mathbf{C}$. Use the canonical basis in the followings.

a) Suppose we have a matrix M which induces a linear map $v \mapsto Mv$. Define that $\lambda \in \mathbf{C}$ is an eigenvalue and $v \in V$ is a corresponding eigenvector.

b) Write the definition that M is upper triangular by denoting the ij -entry of M by m_{ij} and

c) Suppose M is upper triangular and we have a non-zero vector $v \in V$ such that $Mv = 0$. Then, show that $m_{ii} = 0$ for some $i \in \{1, \dots, n\}$.

d) Suppose M is upper triangular and λ is an eigenvalue of M . Then, show that $\lambda = m_{ii}$ for some $i \in \{1, \dots, n\}$.

Problem 4. Let V be a finite dimensional inner product space over $\mathbf{C}$. The norm is induced by the inner product.

a) Write the definition that two vectors $u, v \in V$ are orthogonal ($u \perp v$).

b) Suppose $u \perp v$. Then, show that

$$\|u + v\|^2 = \|u\|^2 + \|v\|^2.$$

c) By using the Cauchy-Schwartz inequality (you don't have to prove it) prove that

$$\|u + v\| \leq \|u\| + \|v\|$$

for all $u, v \in V$.

d) Let $E = \{e_1, \dots, e_n\}$ be a basis for V . Write the definition that E is an orthonormal basis for V .

e) Let $V = \mathbf{C}^3$ and

$$f_1 = (1, 1, 1), \quad f_2 = (0, 1, 1), \quad f_3 = (0, 0, 1).$$

Apply the Gram-Schmidt process to the basis $\{f_1, f_2, f_3\}$ to obtain an orthonormal basis.

Problem 5. let V be a finite dimensional inner product space over $\mathbf{C}$ and $T \in \mathcal{L}(V)$.

a) Write the definition of T^* the adjoint operator of T .

b) Write the definition that T is normal.

c) Prove that T is normal if and only if

$$\|Tv\| = \|T^*v\| \quad \text{for all } v \in V$$

Here, the norm is induced by the inner product. Use the fact that for $S \in \mathcal{L}(V)$ if $\langle Sv, v \rangle = 0$ for all $v \in V$ then $S = 0$.

d) Let $n \times n$ matrix M be

$$m_{kl} = \begin{cases} 2k - 3li & k > l \\ 5 & k = l \\ 2l + 3ki & k < l \end{cases}$$

Here, m_{kl} is the kl -element of M . Show that M is normal.